

Kundai Farai Sachikonye

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🌐 github.com/fullscreen-triangle



Computer-vision engineer: image analysis, GPU image processing, and CNNs — microscopy and camera-based vision, from research to deployed tool.

Profile

Computer-vision researcher and engineer working on image analysis and GPU image processing — segmentation, deconvolution, registration, and morphometry for microscopy, and camera-based vision for physiological signals. I build and deploy validated, GPU-accelerated tools.

Technical Skills

Image analysis	Segmentation, point-spread-function deconvolution, registration, morphometry; scale / coordinate (geometry) reconstruction; uncertainty quantification
Deep learning for vision	PyTorch (CNNs, attention); representation learning; image classification
GPU image processing	WebGL2 / GLSL shaders, WebGPU; real-time image pipelines
Modalities	Fluorescence and phase-contrast microscopy; camera-based (RGB video) vision
Programming	Python (primary) & PyTorch, Rust, TypeScript / WebGL, GLSL, Bash

Experience

2021–present	Independent Computer-Vision & Software Researcher, Bitspark GmbH Built and deployed GPU-accelerated image-analysis and camera-based vision tools; open-source and documented.
2019–2021	Computational Science — ML & HPC, Technische Universität München, Munich CNN-based classification (PyTorch) at HPC scale; large-scale data processing. Taught at Master's level.
2011–2014	Fluorescence Microscopy (BSc thesis), Jacobs University Bremen DNA interstrand crosslinking — cell culture and fluorescence imaging.

Selected Projects

hieronymus	GPU-accelerated microscopy image analysis — segmentation, deconvolution, registration, morphometry; WebGL / GLSL. hieronymus-omega.vercel.app
BRUT (rPPG)	Camera-based remote photoplethysmography — recovers physiological signals (melanin, haemoglobin, vasodilation) from RGB video via a layered optical skin model. remote-photoplethysmography.vercel.app
Microscopy Image Calculus	Formal framework for quantitative microscopy image analysis (deconvolution, scale fields, metric-preserving reconstruction).

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*

2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*

2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*

Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

English Native / Fluent (C2)

German Conversational

resident in Germany since 2011